Experiment No. 4

Verification of System Properties: Linearity, Time Invariance, Causality and Stability

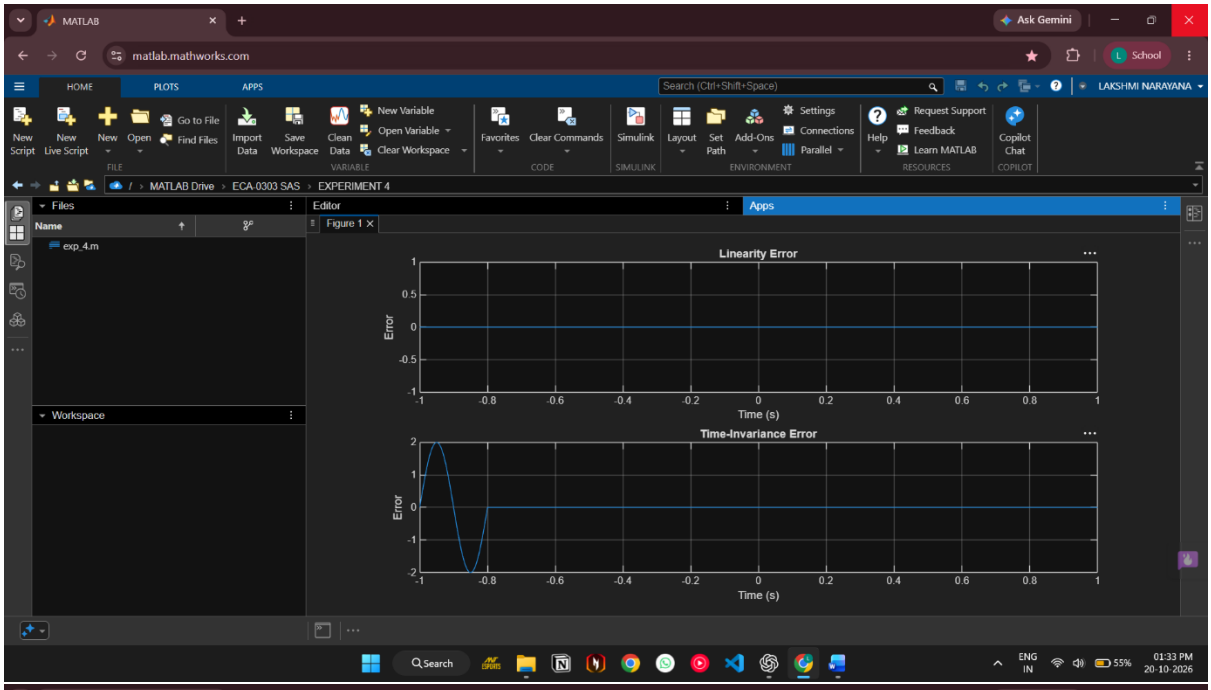
Experimental Procedure

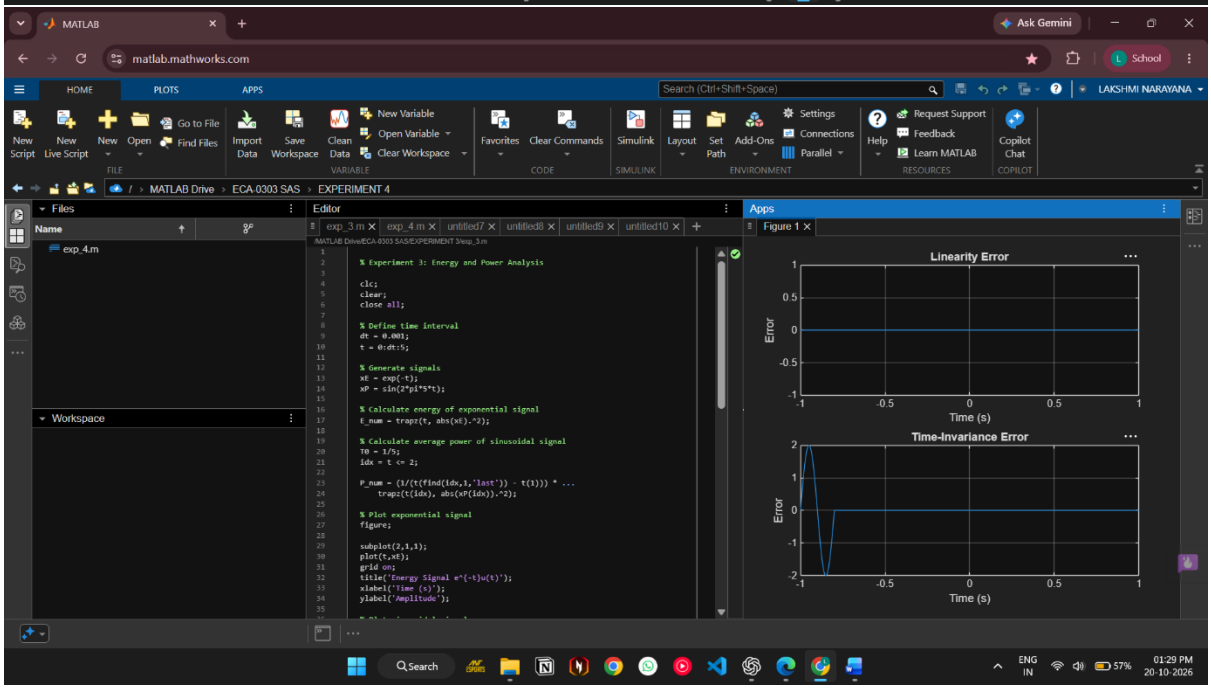
1. Select a representative CT or DT system.
2. Generate two independent test inputs.
3. Compute the individual system outputs.
4. Form a combined input and compare its system output with the corresponding linear combination of outputs.
5. Delay the input by t0 and compare the system response with the correspondingly delayed original output.
6. Calculate numerical differences and classify the system.

MATLAB Coding

```
% Experiment 11: Linearity and Time Invariance
clc; clear; close all;
t = -1:0.001:1;
x1 = sin(2*pi*5*t); x2 = cos(2*pi*5*t);
a = 2; b = 3; t0 = 0.2;
T = @(x) 2*x;
lhs = T(a*x1+b*x2);
rhs = a*T(x1)+b*T(x2);
linError = max(abs(lhs-rhs));
xDelay = sin(2*pi*5*(t-t0));
lhsTI = T(xDelay);
rhsTI = T(x1);
rhsTIshift = interp1(t,rhsTI,t-t0,'linear',0);
tiError = max(abs(lhsTI-rhsTIshift));
figure;
subplot(2,1,1); plot(t,lhs-rhs); grid on; title('Linearity Error');
subplot(2,1,2); plot(t,lhsTI-rhsTIshift); grid on; title('Time-Invariance Error');
fprintf('Linearity error = %g\n',linError);
fprintf('Time-invariance error = %g\n',tiError);
```

Output:


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Figure 1
Linearity Error
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Time-Invariance Error
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MATLAB Drive/ECA-0303 SAS/EXPERIMENT 3/exp_3.m
% Experiment 3: Energy and Power Analysis
clc;
clear;
close all;
% Define time interval
dt = 0.001;
t = 0:dt:5;
% Generate signals
xE = exp(-t);
xP = sin(2*pi*5*t);
% Calculate energy of exponential signal
E_num = trapz(t, abs(xE).^2);
% Calculate average power of sinusoidal signal
T0 = 1/5;
idx = t <= 2;
P_num = (1/(t(find(idx,1,'last')) - t(1))) * ...
trapz(t(idx), abs(xP(idx)).^2);
% Plot exponential signal
figure;
subplot(2,1,1);
plot(t,xE);
grid on;
title('Energy Signal e^{-t}u(t)');
xlabel('Time (s)');
ylabel('Amplitude');
Apps
Figure 1
Linearity Error
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Time (s)
Time-Invariance Error
Error
Time (s)
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